Yacoub Derradji

London, ON | yderradj@uwo.ca | yacoubderradji.dev | linkedin.com/in/yacoubderradji | github.com/ycouby77

Summary

Third-year Mechatronic Systems Engineering student (dual in AI Systems Engineering) who designs robotic mechanisms and takes them toward production: first-principles analysis, SolidWorks and Onshape CAD, ISO 2768 drawings, vendor sourcing, and hands-on prototyping. Also brings ROS 2 integration and Python tooling from autonomous-vehicle work. Interested in robotics, automation, and advanced manufacturing.

Education

Western University Sept. 2024 – Apr. 2029 (Expected)
BESc, Mechatronic Systems Engineering, Dual in AI Systems Engineering *London, ON*

- Cumulative GPA: **3.8 (Dean's Honour List)**
- Coursework: Manufacturing Processes, Mechanics of Materials, Dynamics, Heat Transfer, Fluid Mechanics, Computational Methods, Algorithms & Data Structures

Experience

Research Assistant May 2026 – Present
Advanced Robotics & Mechatronics (ARM) Lab, Western University *London, ON*

- Redesigned a miniaturized planetary-gear magnetorheological (MR) clutch, the per-joint actuator of a multi-finger robotic hand, toward manufacturability
- Traced a torsional shaft fracture and an unmanufacturable keyway to a shared stress-concentration root cause; resolved both by upsizing the shaft (1 to 2 mm) and adopting a D-profile shaft/bore coupling
- Produced the full SolidWorks manufacturing drawing set under ISO 2768-m, catching and correcting a bearing-seat fit error during design review
- Ran multi-vendor sourcing across five manufacturing platforms, securing approved quotes for primary parts

Planning and Controls Developer Oct. 2025 – Dec. 2025
Western Engineering Autopilot (WEAP) *London, ON*

- Cut full-system ROS 2/Gazebo simulation startup by ∼85–90% (3 min → 30 s) with a structured launch architecture for the planning and control subsystems
- Integrated planning and control packages through coordinated topic publishing and deterministic startup sequencing for repeatable testing

Electrical Team Member Sept. 2024 – Mar. 2025
Western Formula Racing (FSAE) *London, ON*

- Created the tractive system schematic and tractive system connector pinouts for the team's electric race car

Projects

Humanoid Robot Neck | *Linkage Design, Onshape, Prototyping* Summer 2026

- Designed a compact 2-DOF neck for Baker Bot, a torso-up humanoid (WE First club): two 40 kg servos drive ball-jointed pushrods against a fixed-length link to produce pitch and roll
- Owned CAD (off-the-shelf rod ends, custom 3D-printed mounts), bench prototyping, and fabrication; integrated the neck on the robot and handed off to the software and torso subteams

Off-Road RC Vehicle | *SolidWorks, Mechanism Design, Embedded Control* Dec. 2025 – Apr. 2026

- Designed the full vehicle in SolidWorks (chassis, modular FWD/4WD drivetrain, steering, suspension) and built it from mostly 3D-printed parts under size and budget limits, with no kit
- Integrated Arduino control with motor drivers and servo steering; revised suspension arm lengths and steering knuckles in a second design pass

Tendon-Driven Robotic Gripper | *Tendon Actuation, Grasping* Sept. 2025 – Present

- Built a 3-finger gripper prototype, adapting an open-source finger design for rubber-band and string tendons, with a 3-servo palm and one finger opposing two

Skills & Certifications

CAD & DFM: SolidWorks (CSWA), Onshape, GD&T, ISO 2768 Drawings, DFM, Shaft/Coupling & Bearing Design, Gear Trains, 3D Printing
Robotics & Simulation: ROS 2, Gazebo, MATLAB, Simulink, Linkage Design, Tendon Actuation, Servo Control
Electrical & Embedded: Arduino, ESP32, Motor Drivers, PWM Control, Schematics & Pinouts, Soldering
Programming: Python, C/C++, Java, Git/GitHub

Certifications & Awards: Certified SolidWorks Associate (CSWA); Dean's Honour List; 3rd Place, Western Engineering Competition, Junior Division (Nov. 2025); Top 3 of 72 Teams, Ivey Case Competition (Mar. 2025)